

Oral Assessment

Group EDA Recap

Dataset: Steam Games Dataset (Kaggle, published by Artemiy Ermilov)

Cleaned March 2025 version, 89,618 games, 47 variables, from April 1997 to March 2025

Research Question

S1	What makes a popular game on Steam?
S2	What is the relationship between a game genre and its number of positive reviews?
S3	What is the relationship between a game genre and its proportion of positive reviews? (after accounting for price and age rating)

Key EDA findings

- num_reviews_total and pct_pos_total are both set to -1 for games with fewer than 10 reviews
- pct_pos_total and num_reviews_total are set to -1 for games with under 10 reviews.
- Group applied a 30-review minimum, keeping 34,583 of 89,618 games (38.6%).
- 109 games with no genre listed were dropped from the review-filtered set.
- Six non-genre labels excluded (Violent, Gore, Nudity, Sexual Content, Free to Play, Early Access).
- This brought 27 raw genre tags down to 21 genres.
- A 100-games-per-genre minimum left 10 established genres.
- Positive review share by genre: Casual highest at 83% median, Massively Multiplayer lowest at 67%.
- Price by genre: Sports and RPG highest at \$7.99 median, Massively Multiplayer near \$0 (free-to-play).
- Age restriction share by genre: Action highest at 3.9%, Casual lowest at 0.5%.
- Price and age-restriction show some association with review positivity across genres
- This confounding pattern directly motivates my Phase 2 question.

EDA Summary

We filtered to games with at least 30 reviews, which kept about 39% of the dataset, since games under 10 reviews have their review stats masked as -1 by Steam. After dropping ungenred games and non-genre tags like Early Access or Gore, we had 21 genres, and after a 100-games-per-genre minimum, 10 established genres remained. Positive review share ranges from 83% median for Casual down to 67% for Massively Multiplayer. Price and age-restriction share vary a lot by genre too - RPG and Sports sit around \$8 median, Massively Multiplayer is close to free. That variation is what's motivating my Phase 2 question: whether genre still predicts positive review share once you control for price and age rating.

Proposed Analysis Plan

- **Response variable:** pct_pos_total (positive review proportion), logit transformed
- **Predictors:** primary_genre (categorical), price (continuous, possibly log), required_age (categorical/ordinal)
- **Model:** $\text{logit}(\text{pct_pos_total}) \sim \text{genre} + \text{price} + \text{required_age}$, OLS on transformed scale n or GLM if needed
- **Structure:** ANCOVA-style, genre as the factor of interest, price and age as covariates
- **Filtering:** same 30-review and 100-games-per-genre thresholds as the group EDA
- **Significance:** overall F-test, plus t-tests/contrasts on genre coefficients vs. the reference genre
- **Scope:** one algorithm, one response variable, small, justified predictor set, per the marking guidance

Points to Raise with Frery

- Confirm primary_genre is right, since group EDA used exploded genres
- Is logit + OLS fine, or push to GLM/beta for regression?
- Do group thresholds still hold for individual regression?
- Hypothesis tests only, or fitted model with goodness-of-fit too?